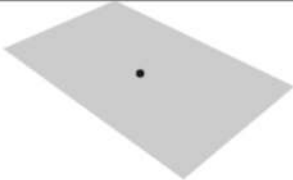
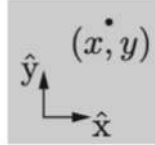
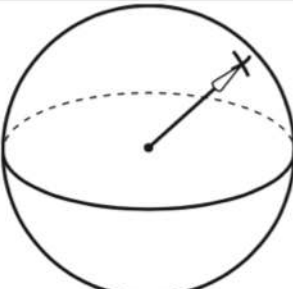
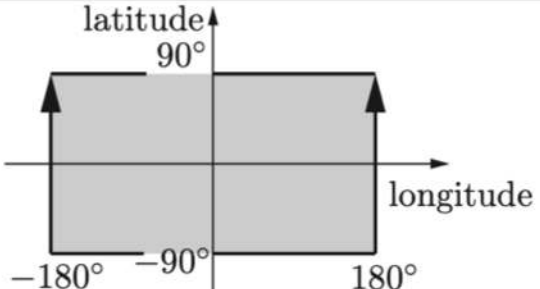
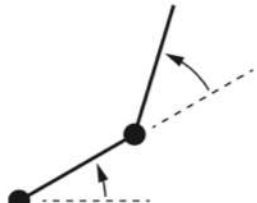
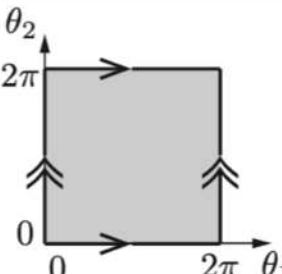
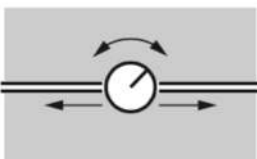
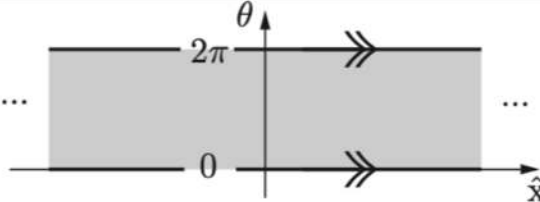


Chapter 2	Configuration Space
	2.1 DOF of a Rigid Body
	2.2 DOF of a Robot
	2.3 C-space Topology and Representation
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- Two C-spaces may have the same dof but differ in other ways. The **topology** (“shape”) of a space is independent of how we **represent** it.

system	topology	sample representation
 point on a plane	 $\mathbb{E}^2$	 $\mathbb{R}^2$
 spherical pendulum	 S^2	 $[-180^\circ, 180^\circ] \times [-90^\circ, 90^\circ]$
 2R robot arm	 $T^2 = S^1 \times S^1$	 $[0, 2\pi) \times [0, 2\pi)$
 rotating sliding knob	 $\mathbb{E}^1 \times S^1$	 $\mathbb{R}^1 \times [0, 2\pi)$

Any value in an atlas of coordinate charts?
 An implicit representation?

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 An implicit representation?

Any value in an atlas of coordinate charts?
 An implicit representation?



hexrotor with two 5-DOF arms

<https://www.prodrone.com/archives/1420/>

C-space topology, with and without
arm joint limits, rotor angles?
Implicit/explicit representations?
Grübler's formula?



C-space topology and representation?
Include gripper, wheel angles?

KUKA youBot
mecanum-wheel omnidirectional base
moving on flat ground
plus 5-DOF robot arm + gripper